

Congestion-aware search

Lee ignores other nets

The idea

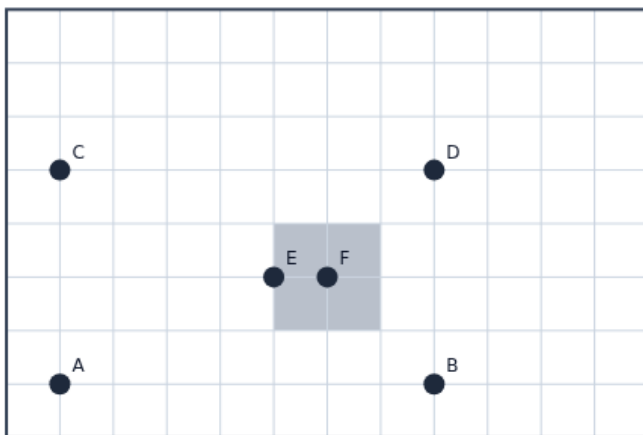
- Maintain open heap ordered by f equals g plus Manhattan distance to goal
- Expand four neighbors on M1 for horizontal steps and M2 for vertical
- Step cost is one plus ten times overflow penalty when usage meets capacity
- First time you pop the goal, return the path

Track cost

A* ROUTE

Track cost

Step 1 / 5 · cost · module02-03-astar-route



Explanation

A* adds penalty when usage $\geq$ capacity.

- Manhattan h
- Detour allowed

Penalize hot tracks

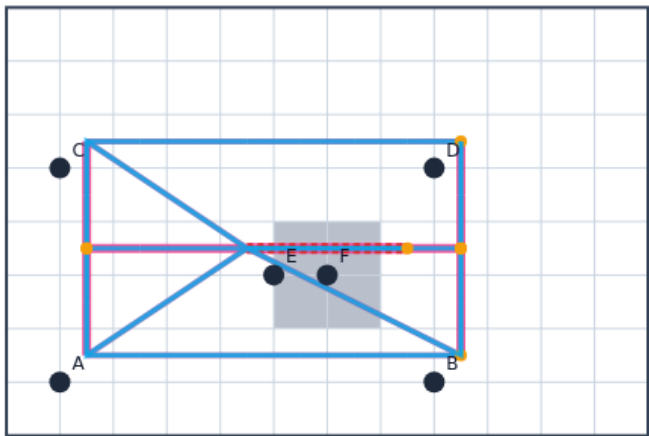
Track cost

Layered overflow

A* ROUTE

Layered overflow

Step 2 / 5 · l-ov · module02-03-astar-route



Explanation

Layered L-HV on spread overflows shared tracks.

- Shared M1/M2

total=3

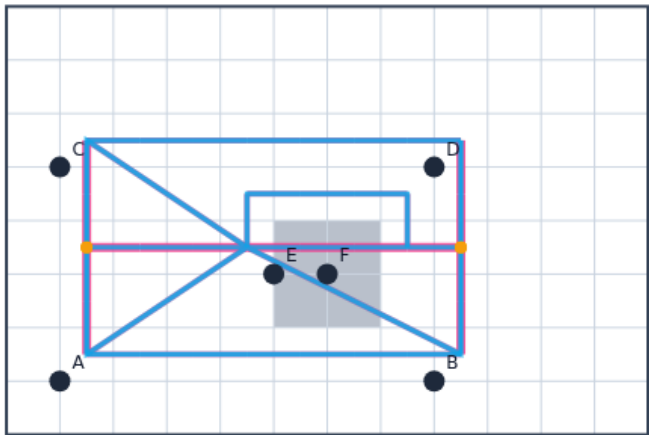
Layered overflow

Route A*

A* ROUTE

Route A*

Step 3 / 5 · astar · module02-03-astar-route



Explanation

Sequential A* considers current usage.

- Six nets

total≈0

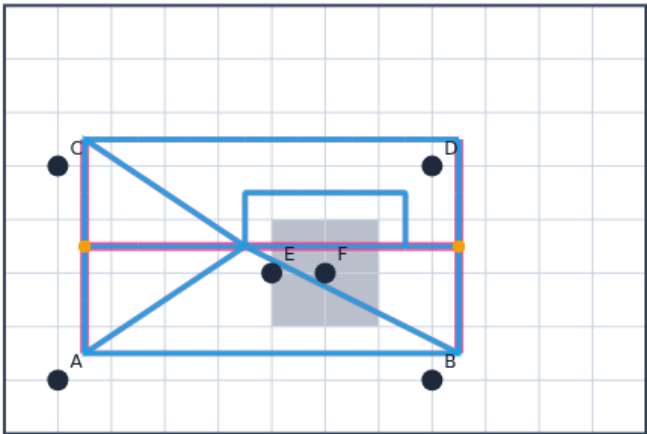
Route A*

Compare totals

A* ROUTE

Compare totals

Step 4 / 5 · compare · module02-03-astar-route



Explanation

A* may clear overflow on the toy spread.

- Not magic

Still Cap=2

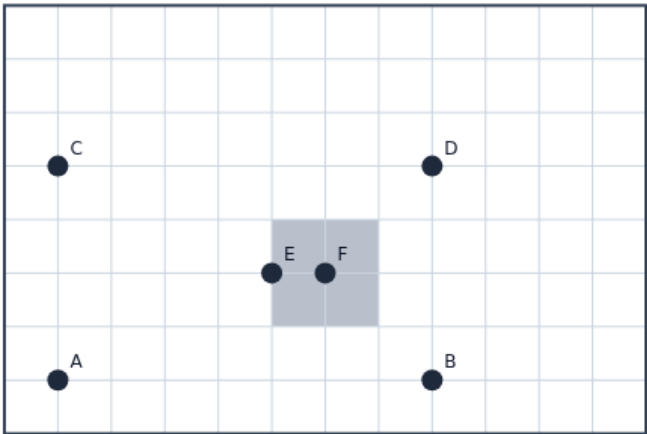
Compare totals

Move pins

A* ROUTE

Move pins

Step 5 / 5 · move · module02-03-astar-route



Explanation

Moving cells changes pin grid and paths.

- Learner state

Check scores routes

Move pins

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / A* detailed routing

INTERACTIVE TOOL

A* detailed routing

A* penalizes saturated tracks; may beat layered L overflow on spread starter.

Your job: route layered L first (overflow ≈ 3), then A* (golden total ≈ 0).

Reset workspace

Challenges0 / 8 cleared

Pick one

Intro

Layered has overflow

Pick a challenge and click **Start**. Move cells, route, then **Check**.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Route Lee

Route layered

Route A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0

Browser lab starter

learn_routing — 03 astar route

Implement track

- Implement or call ``astar_route(start, goal, usage, cap, nx, ny, blocks)``
- Block the hot edge and show A* cannot go direct; unblock a vertical detour path exists

Pitfalls

- Using undirected track keys inconsistently
- Checking capacity on the wrong layer for a move
- Forgetting blockages still forbid cells even when tracks are free

Your turn

- Clear A* challenges
- Next: deposit segments on tracks and measure overflow